

Incorporating Multispecies Interactions into Stock-Recruitment Modelling

MMath Extended Independent Project

Alice Woodley

Abstract

Recruitment, the process by which young fish mature and grow to a specified age, is an important consideration for fishery scientists when conducting stock assessments. The number of recruits in a given year is linked to the size of the fish stock and such a relationship can be modelled mathematically to help assess the sustainability of the stock in future years. However, stock-recruitment data can appear almost random and fitting a mathematical relationship is often unsuitable for most species. Additional explanatory variables, such as sea temperature, have been added to stock-recruitment models in an attempt to better explain variation in the data. As multispecies interactions have a clear effect on species abundance, here we adapt this approach to incorporate this effect into a stock-recruitment model by considering the population size of managed predator and prey species. Stock assessment data of six species from the Gulf of Alaska is used to fit standard and multispecies Ricker models via non-linear regression, which are compared through the Akaike Information Criterion. After fitting three multispecies models, questions remain around the validity of the multispecies models for the Pacific ocean perch and walleye pollock. We found that the inclusion of the stock size of some prey species of Pacific cod did make an improvement on the standard stock-recruitment model. Limitations of this method restrict its practicality in stock assessments. Nevertheless, considering species interactions could improve the accuracy of recruitment predictions, which in turn supports confident decision-making to maintain an important global food source.

Contents

1	Introduction	2
1.1	Project aim	2
1.2	Summary of contents	3
2	Biological and mathematical background	4
2.1	Recruitment and its relationship with stock size	4
2.1.1	Beverton-Holt model	4
2.1.2	Ricker model	5
2.2	Incorporating additional effects	6
3	Data and methods	7
3.1	Data selection	7
3.2	Statistical methods	10
3.2.1	Non-linear least squares regression	10
3.2.2	Bootstrap resampling for confidence intervals	11
3.2.3	Akaike Information Criterion	12
3.3	Applications to standard and multispecies Ricker models	12
4	Results and discussion	14
4.1	Single-species Ricker models	14
4.2	Multispecies Ricker models	16
4.2.1	Pacific ocean perch model	16
4.2.2	Walleye pollock model	18
4.2.3	Pacific cod model	20
5	Conclusion	23
5.1	Summary	23
5.2	Project limitations and future work	24
	Appendix - an unusual year for walleye pollock	26
	References	28

1 Introduction

Fish make up 15% of global animal protein consumption, a figure that maintains the livelihoods of over 61 million people around the world [1]. Although world hunger has decreased in recent years, it is estimated that 512 million people will still be facing hunger in 2030 [2]. Therefore, sustainable fisheries management is vital not only in the conservation of aquatic species during a time of unprecedented environmental change [3], but also in maintaining a key source of food for a growing global population.

Each year, fisheries carry out stock assessments (e.g. [4]) to evaluate the effect current fishing practices are having on their stock and to investigate how these practices could impact the stock in future. New data is collected and used alongside historical catch data and estimates from previous stock assessments to fit a mathematical model, which is then able to predict future values of key variables. Outputs from the model are then used to calculate the overfishing limit (OFL), the acceptable biological catch (ABC) and the annual catch limit (ACL) for that year. These statistics inform decision-making in management of the fishery and the annual catch limits are communicated directly to the fishermen to guide their work.

Models used in stock assessments often find a balance between complexity and ease of use, so scientists can easily communicate the results to managers and fishermen [5]. For this reason, the specific choice of model can depend on the fishing region, the managed species, the amount of data available, and simply personal preference of fishery scientists. Such models are usually age-structured and almost always consider only the single species of interest. However, with improvements in computing power and statistical software, more detailed models are becoming easier to use. Fishery scientists have now been able to consider adding additional effects into models to improve their biological accuracy without compromising on convenience. For example, environmental variables, such as sea temperature, are known to have an effect on species behaviour, which in turn has known consequences for fisheries [6]. This effect is becoming increasingly more important to consider as we learn more about the effects of climate change [7].

Interactions between species, particularly in relation to the marine food web, have an obvious impact on some values estimated in stock assessments, as an increase in predator abundance will directly cause a decrease in prey abundance. The desire to incorporate this effect into stock assessments has driven recent mathematical research: existing models have been adapted to include multiple species [8], and entirely new models have been created specifically for this purpose [9]. Despite this, these multispecies models are only occasionally used in stock assessments, when standard models are unable to accurately estimate a particular value due to a lack of data. This is likely due to their increased complexity over ease of use relative to single-species models.

1.1 Project aim

The size of the fish stock is arguably the most important estimated value in a stock assessment, along with the number of new adult fish joining the fishery through the recruitment process. These two values are known to be related in some way, and this relationship is vital to consider within a stock assessment.

This project aims to incorporate the effect of predator-prey interactions between managed

fish species into a multispecies model of the relationship between stock size and recruitment. To achieve this, we adapt the existing ideas behind adding additional explanatory variables to stock-recruitment models to include the abundance of interacting species. Such a model will then be compared to a traditional single-species model to evaluate whether or not the inclusion of this effect has improved the model.

1.2 Summary of contents

In this project report, our research is structured as follows:

- In section 2, the background of the project is explained in depth, defining the recruitment process, exploring the mathematics behind stock-recruitment relationships, and deriving the model that is used throughout the rest of the project.
- In section 3, the data used in this project and the species in question are introduced. We then outline the practical methods to be used to fit models, interpret them and compare them.
- In section 4, the results of applying these methods to the chosen data are presented.
- In section 5, we conclude by summarising the project and its results, discussing its limitations and possible areas of future work.

2 Biological and mathematical background

2.1 Recruitment and its relationship with stock size

“The most important and generally most difficult problem in biological assessment of fisheries is the relationship between stock and recruitment... Any fishery manager who acts as if recruitment will remain constant as a fishery increases is foolish.” - Hilborn and Walters, 1992 [10]

Recruitment is the process by which young fish, as eggs and larvae, transition to an older life stage. Usually at the life stage they are first able to reproduce, they are considered recruits and have been recruited to the fishery [11].

From this definition, it is clear that the size of a fish stock has a relationship with the number of recruits in a given year. Attempts have been made to model this relationship mathematically, but questions remain around whether such a relationship is significant and whether there is an optimal model [12]. The severity of this challenge stems from the fact that stock-recruitment data can seem almost random and frequently has no visible trend. Very few species show a strong relationship and even then, fishery scientists most often choose to ignore any possible relationship and estimate recruitment as a deviation from the mean (e.g. [4]).

Additionally, different measures of stock size can be used in such a model, for example, number of eggs [10], but the choice of measurement can impact model fit. In this project, we measure the stock size as the total mass of all fish at reproductive age, referred to as the spawning stock biomass (SSB), measured in tonnes.

2.1.1 Beverton-Holt model

The mathematical intuition behind some of the most popular stock-recruitment models is summarised by Quinn and Deriso [13].

Let Z_t represent the mortality of a species at age t . Population dynamics can then be represented by the differential equation:

$$\frac{dN}{dt} = -Z_t N, \quad (1)$$

where N is the total number of fish in the population and $N(0) = N_0$ is therefore the total number of eggs (fish at age 0). Now suppose Z_t is influenced by effects that are either dependent or independent of population density. This can be written as:

$$Z_t = a_1 + b_1 N, \quad (2)$$

which can then be substituted into equation (1):

$$\frac{dN}{dt} = -(a_1 + b_1 N)N.$$

This equation has the solution:

$$N(t) = \frac{N_0}{e^{a_1 t} - \frac{b_1}{a_1} N_0 + \frac{b_1}{a_1} N_0 e^{a_1 t}}. \quad (3)$$

By assuming the number of eggs produced N_0 is related to the size of the spawning stock S and fecundity f (a measure of fertility):

$$N_0 = fS, \quad (4)$$

and by fixing $t = T$ where T is the age of recruitment, equation (3) becomes:

$$N(T) = R = \frac{S}{\frac{e^{a_1 T}}{f} + \frac{b_1}{a_1}(e^{a_1 T} - 1)S},$$

which can be reparametrised for simplicity:

$$R = \frac{\alpha S}{1 + \beta S}. \quad (5)$$

Equation (5) is the Beverton-Holt spawner-recruit curve [14], where α is the density-independent productivity parameter, and β indicates the level of density-dependence.

Figure 1 (a), found in the following subsection, gives an example of the shape of the Beverton-Holt curve.

2.1.2 Ricker model

Alternatively, assume that mortality is influenced by spawning stock so the young fish population is inhibited in some way, for example, through cannibalism or competition for resources. Equation (2) is instead written as:

$$Z_t = a_2 + b_2 S. \quad (6)$$

As before, this can be substituted into equation (1) to give:

$$\frac{dN}{dt} = -(a_2 + b_2 S)N,$$

which can be solved to find:

$$N(t) = N_0 e^{-(a_2 + b_2 S)t}.$$

Again, by fixing $t = T$ and substituting equation (4), we find:

$$N(T) = R = (f e^{-a_2 T}) S e^{-b_2 T S},$$

which can be simplified:

$$R = \alpha S e^{-\beta S}. \quad (7)$$

Equation (7) is the Ricker spawner-recruit curve [15], where parameters α and β are defined as in the Beverton-Holt model. Figure 1 (b) gives an example of the curve shape produced by a Ricker model.

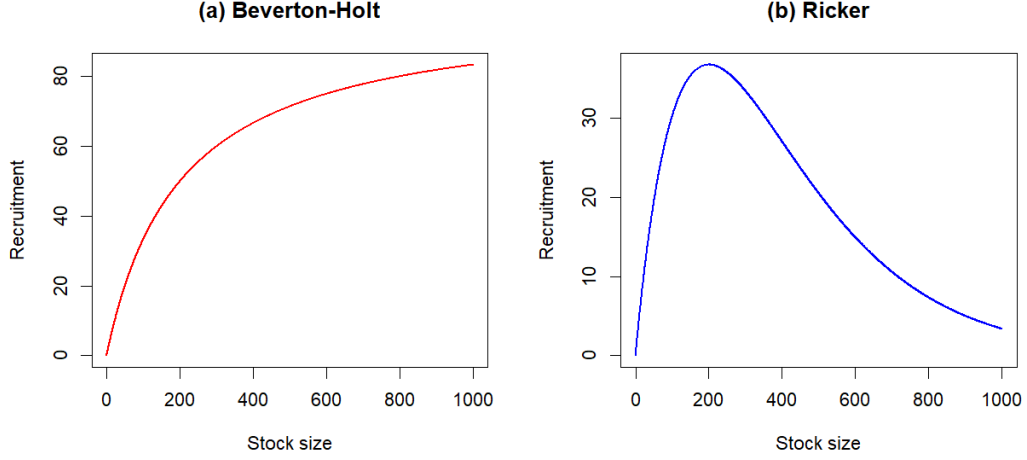


Figure 1: Examples of (a) Beverton-Holt and (b) Ricker curves. Both of these examples use $\alpha = 0.5$ and $\beta = 0.005$.

Note in this case, as $S \rightarrow \infty$, $R \rightarrow 0$. This indicates that at a certain point, a large population size has a detrimental effect on recruitment numbers, due to our consideration of an inhibiting effect.

2.2 Incorporating additional effects

In attempts to improve the fit of stock-recruitment models, other possible effects on fish during the recruitment process have been considered. Suppose other variables of interest are $X_1, \dots, X_p$. Returning to equation (6), we can add in these variables to give:

$$Z_t = a_2 + b_2 S + c_1 X_1 + \dots + c_p X_p,$$

which, following the same steps, results in the generalised Ricker function:

$$R = \alpha S e^{-\beta S + c_1 X_1 + \dots + c_p X_p}.$$

This approach has mainly been used to consider environmental variables, such as sea temperature [16], but here we will use this generalised form to incorporate the effects of species interactions.

3 Data and methods

3.1 Data selection

The data used in this project is sourced from the RAM Legacy Stock Assessment Database [17], which contains data from around 1500 fish stocks globally (as of version 4.66).

Ideally, a fishing region should be selected from the database that has SSB and recruitment time series available for more than two species. By enforcing this criteria, there were 132 regions to choose from. The region must also have sufficiently long time series over the same time period for each species. We also narrow the search by considering the effect of interspecies interaction through predator-prey relationships, as the effect species have on their respective population sizes is intuitive. For example, decreased abundance of a prey species would lead to a decreased abundance of predators due to an insufficient source of food. We specify that the selected region should not have too many interacting species (< 10) for simplicity. Keeping these criteria in mind, manual research led to the choice of the Gulf of Alaska. A map of the fishing boundaries can be found in figure 2.

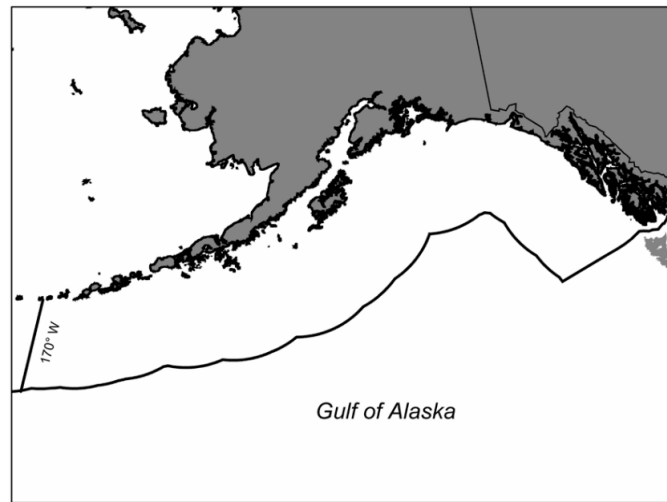


Figure 2: Boundaries of Gulf of Alaska fisheries [18].

The Gulf of Alaska has 19 species present in the RAM Legacy database, all managed by the National Oceanic and Atmospheric Administration (NOAA). However, only 12 have SSB and recruitment time series available. Through research in the literature, six of these were determined as neither predators nor prey of any other species in the group of 12. This leaves a group of six species to use in the stock-recruitment models:

- Juvenile arrowtooth flounder (*Atheresthes stomias*) are considered recruits at age 1.
- Young flathead sole (*Hippoglossoides elassodon*) are considered recruits before they reach age 1.
- Young Pacific cod (*Gadus macrocephalus*) are also considered recruits before they reach age 1.

- Juvenile Pacific ocean perch (*Sebastes alutus*) are considered recruits at age 2.
- Young rock sole (*Lepidopsetta bilineata*) are considered recruits before they reach age 1.
- Juvenile Alaska pollock (*Gadus chalcogrammus*), referred to as walleye pollock in the database and throughout this project, are considered recruits at age 1.

Note that recruitment ages can vary in the literature, sometimes provided as a range of ages rather than a single year. The recruitment ages considered in this project are those recorded in the database [17].

Figure 3 illustrates the predator-prey interactions among these six species ¹ [20–22] .

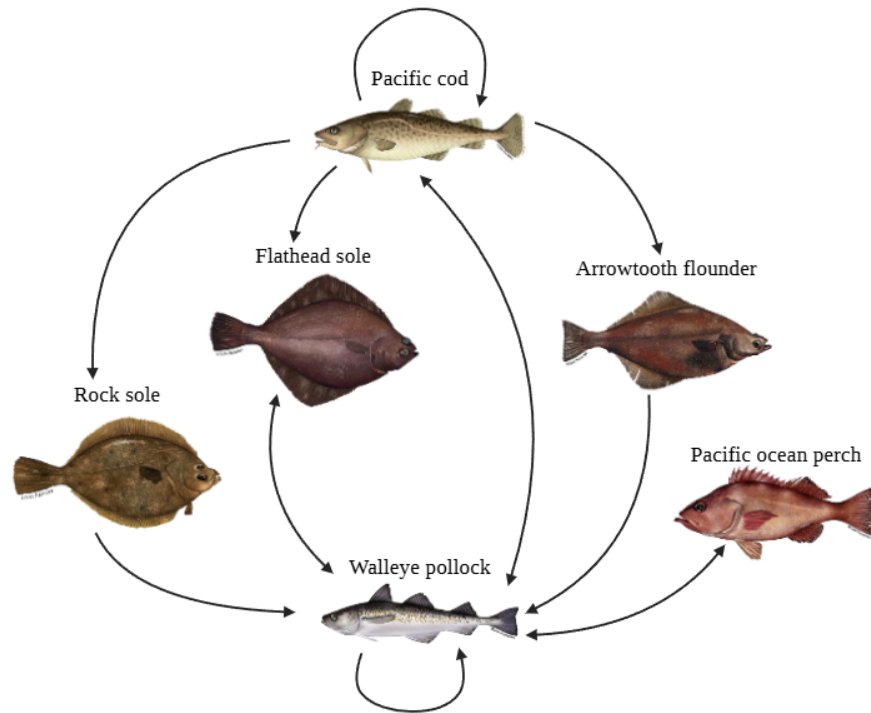


Figure 3: Predator-prey interactions between Gulf of Alaska species ². An arrow pointing from species A to species B indicates species A is a predator of species B.
Note both Pacific cod and walleye pollock are cannibalistic species.

These relationships refer to the predator feeding on the juvenile fish of their prey species, which will impact recruitment. It is also important to note this is not representative of the entire food web: these species have other main food sources and could interact with species that are not managed, and therefore will not necessarily have data recorded.

¹Information used in the diagram was initially found on FishBase [19], who provide the referenced sources.

²This figure was created independently using illustrations from NOAA Fisheries [23].

Figure 4 displays the SSB and recruitment time series of each species. The time series are formed of estimates calculated for yearly stock assessments. These are based on regular acoustic surveys, which estimate abundance using sound waves, and bottom trawl surveys, where a large net is towed along the seafloor. Estimates from previous years are also considered.

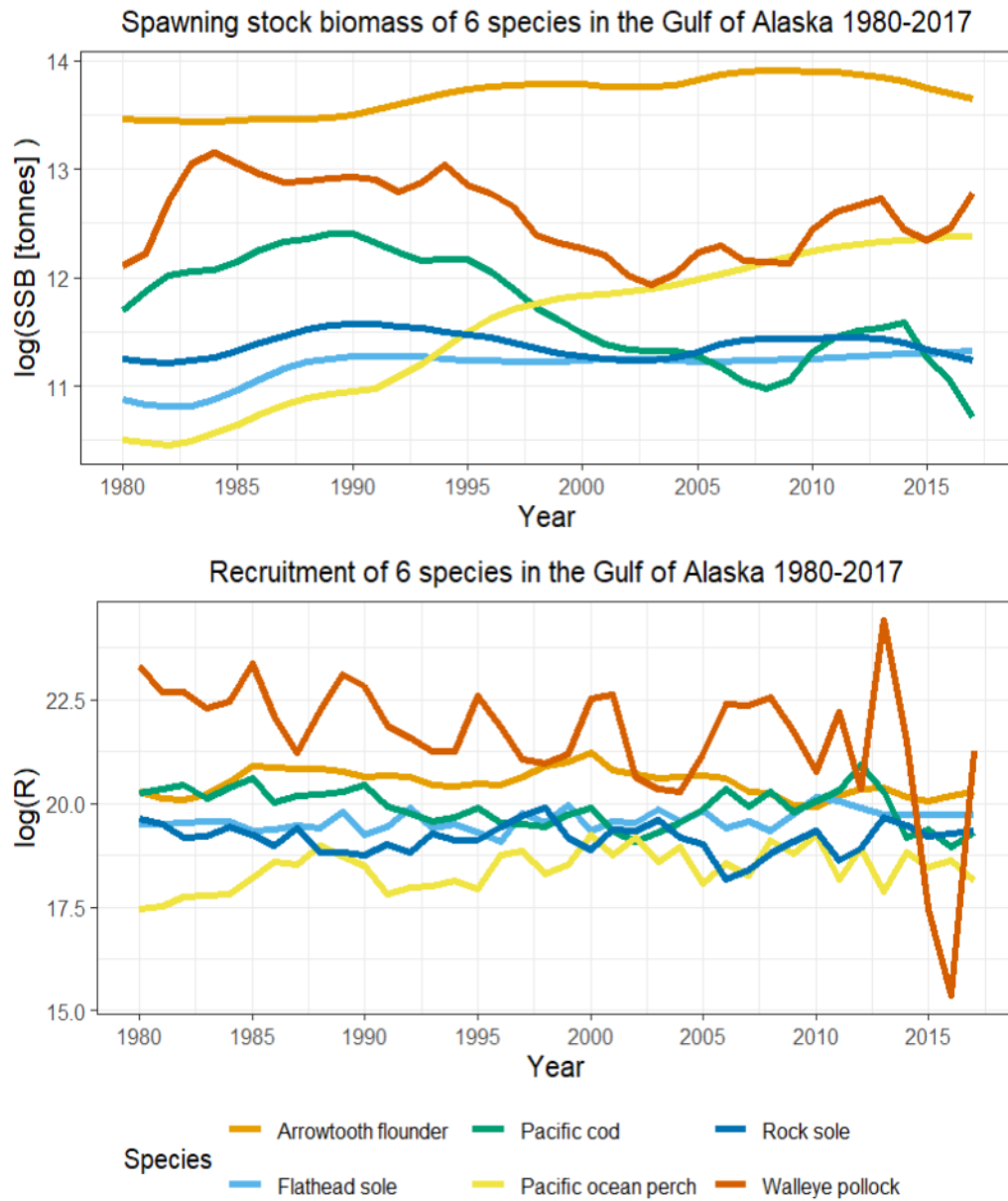


Figure 4: Spawning stock biomass (SSB) and recruitment of selected species in the Gulf of Alaska from 1980 to 2017. Note the logarithmic scale is used here for better comparison between species.

Note the spike in walleye pollock recruitment, which represents the unusually large recruit cohort in 2013 (born in 2012), followed by an unusually small cohort the following year. The reason behind this anomaly is unclear, but the 2013 recruits went on to experience marine heatwaves in the region in 2015-2017, and displayed unusual life history characteristics, such as reduced growth [4]. We continue to include these data points in the models, as they could help highlight the effect of interactions with walleye pollock, especially as it is preyed on by all other species.

Before fitting any models, to take into account the recruitment ages of each species, the recruitment time series were lagged to represent the impact of stock size on recruitment more accurately. For example, as juvenile Pacific ocean perch are recruited at age 2, the spawning stock of 2000 will be responsible for the recruitment in 2002.

3.2 Statistical methods

We now require statistical methods in order to find parameter estimates, interpret them and compare the resulting models.

3.2.1 Non-linear least squares regression

Regression methods often find parameter estimates by minimising the sum of the square residuals: such methods are often referred to as least squares regression and are some of the oldest statistical methods which remain popular even today [24]. The non-linear nature of the models in this project means ordinary linear regression methods will not be appropriate. There are several methods available to deal with non-linearity in least squares estimation. Here, we use the most popular: the Gauss-Newton method, following the description by Bates and Watts [25].

Consider the non-linear regression model for response variable $\mathbf{Y}$ given covariates $\mathbf{x}$:

$$\mathbf{Y} = f(\mathbf{x}, \boldsymbol{\beta}) + \boldsymbol{\varepsilon},$$

where f is the model function, $\boldsymbol{\varepsilon}$ is a random error, typically following a normal distribution, and $\boldsymbol{\beta}$ is a vector of parameters to be estimated. Gauss proposed using a linear approximation of f to iteratively improve an estimate for $\boldsymbol{\beta}$ starting with an initial guess $\boldsymbol{\beta}^0$. The linear approximation of f is found by calculating the first order Taylor series of f about $\boldsymbol{\beta}^0$:

$$f(\mathbf{x}, \boldsymbol{\beta}) \approx f(\mathbf{x}, \boldsymbol{\beta}^0) + v_1(\beta_1 - \beta_1^0) + v_2(\beta_2 - \beta_2^0) + \cdots + v_p(\beta_p - \beta_p^0),$$

where the coefficients v_i are calculated as:

$$v_i = \left. \frac{\partial f(\mathbf{x}, \boldsymbol{\beta})}{\partial \beta_i} \right|_{\boldsymbol{\beta}^0} \quad \text{for } i = 1, \dots, p.$$

This is equivalent to approximating the residuals $\mathbf{r}(\boldsymbol{\beta}) = \mathbf{Y} - f(\mathbf{x}, \boldsymbol{\beta})$ by:

$$\mathbf{r}(\boldsymbol{\beta}) \approx \mathbf{Y} - [f(\mathbf{x}, \boldsymbol{\beta}^0) + \mathbf{V}^0 \boldsymbol{\delta}] = \mathbf{r}^0 - \mathbf{V}^0 \boldsymbol{\delta},$$

where $\mathbf{r}^0$ are the residuals calculated using $\boldsymbol{\beta}^0$, $\boldsymbol{\delta} = \boldsymbol{\beta} - \boldsymbol{\beta}^0$ and $\mathbf{V}^0$ is a $1 \times p$ vector of the coefficients v_i . We can now update the estimate for $\boldsymbol{\beta}$ by adding the Gauss increment $\boldsymbol{\delta}^0$:

$$\boldsymbol{\beta}^1 = \boldsymbol{\beta}^0 + \boldsymbol{\delta}^0,$$

which we calculate by minimising the residual sum of squares $\|\mathbf{r}^0 - \mathbf{V}^0 \boldsymbol{\delta}\|^2$. Successive iterations are performed using the updated estimate $\boldsymbol{\beta}^1$ to calculate new residuals, a new coefficient vector $\mathbf{V}^1$ and a new Gauss increment $\boldsymbol{\delta}^1$ to update $\boldsymbol{\beta}^1$, and so on. Iterations are repeated until the sequence of estimates converges to an optimal estimate $\hat{\boldsymbol{\beta}}$.

3.2.2 Bootstrap resampling for confidence intervals

Bootstrapping is a method commonly used across statistics to provide insight into the accuracy of calculated estimates by creating a sample of estimates calculated from artificial data. Here, we use bootstrapping to find confidence intervals for the parameters in a non-linear model. Huet et al. [26] describe the construction of bootstrap confidence intervals in the following way.

When performing non-linear regression on the model $\mathbf{Y} = f(\mathbf{x}, \boldsymbol{\beta}) + \varepsilon$, we find an estimate $\hat{\boldsymbol{\beta}}$ of the model parameters using our chosen data. To get a sample of bootstrap estimates $\hat{\boldsymbol{\beta}}^*$, first, samples of artificial data (x_i, Y_i^*) are calculated, where:

$$Y_i^* = f(x_i, \hat{\boldsymbol{\beta}}) + \varepsilon_i^*, \text{ for } i = 1, \dots, k,$$

where k is the number of parameters $\boldsymbol{\beta}_i$. The random error terms ε_i^* are simulated using the residuals of the non-linear regression model:

$$\hat{\varepsilon}_i = Y_i - f(x_i, \hat{\boldsymbol{\beta}}),$$

which are transformed to find centered residuals:

$$\tilde{\varepsilon}_i = \hat{\varepsilon}_i - \hat{\varepsilon}_\bullet,$$

where $\hat{\varepsilon}_\bullet$ is the mean of the residuals:

$$\hat{\varepsilon}_\bullet = \sum_{i=1}^k \frac{\hat{\varepsilon}_i}{n}.$$

We then draw a random sample of n ε_i^* (with replacement, each with probability $\frac{1}{n}$) from the empirical cumulative distribution of the centered residuals $\tilde{\varepsilon}_i$.

$\hat{\boldsymbol{\beta}}^*$ is then calculated as the value of $\boldsymbol{\beta}$ that minimises:

$$C^*(\boldsymbol{\beta}) = \sum_{i=1}^k (Y_i^* - f(x_i, \boldsymbol{\beta}))^2.$$

The sample containing B bootstrap estimates $\{\boldsymbol{\beta}^{*,b} : b = 1, \dots, B\}$ approximates the distribution of $\hat{\boldsymbol{\beta}}$, so we can use this to give us an idea of its accuracy. Here, we wish to calculate confidence intervals for $\boldsymbol{\beta}$, which can be achieved as follows.

Consider the test statistics:

$$\hat{T}^* = \frac{\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}}}{S_{\hat{\boldsymbol{\beta}}^*}} \text{ and } \hat{T} = \frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{S_{\hat{\boldsymbol{\beta}}}},$$

where $S_{\bullet}$ is the standard error of the specified estimate. Traditionally, to find a confidence interval for β , we would find quantiles of the distribution of $\hat{T}$. The difference between $\hat{T}^*$ and $\hat{T}$ tends to 0 when n is large³. Here, we can use quantiles of the distribution of $\hat{T}^*$ in place of $\hat{T}$ to construct confidence intervals as below.

Let $\{\hat{T}^{*,b} : b = 1, \dots, B\}$ be a sample of $\hat{T}^*$ of size B . Let b_{α} be the α percentile of this sample. It is possible to prove that $P(\hat{T} \leq b_{\alpha})$ tends to α when n is large. We can now construct an $\alpha\%$ bootstrap confidence interval for β :

$$\hat{I}_B = [\hat{\beta} - b_{\frac{1-\alpha}{2}} S_{\hat{\beta}}, \hat{\beta} - b_{\frac{\alpha}{2}} S_{\hat{\beta}}].$$

3.2.3 Akaike Information Criterion

We want to be able to compare the new multispecies model to the standard Ricker model. To do this, the Akaike Information Criterion (AIC) can be calculated [28]:

$$\text{AIC} = 2k - 2\log(\hat{L}), \quad (8)$$

where k is the number of parameters in the model and $\hat{L}$ is the maximum likelihood estimate, obtained by maximising the likelihood function:

$$L(\beta | \mathbf{x}) = \prod_{i=1}^n f(x_i | \beta),$$

where f is the probability density function of the random error ε ⁴. There are many ways to compare mathematical models, and often, comparison methods focus solely on the model accuracy. By incorporating the number of parameters in the model, the AIC additionally considers complexity, which will be advantageous here as models for species linked with a large number of predator or prey species will have many more added parameters. Also, for such models to be useful in stock assessments, fishery scientists value ease of use, which is a less common trait of a model with many parameters to be estimated.

3.3 Applications to standard and multispecies Ricker models

To fit models in practice, the above methods were all implemented using R (v4.4.1) [29], guided by the textbook *Introductory Fisheries Analyses with R* by Derek Ogle [30].

All the code used in this project is available in a GitHub repository, which can be found at: <https://github.com/alicedwoodley/Extended-Independent-Project>.

Firstly, a standard Ricker model (equation (7)) will be fitted to the data for each species to evaluate the suitability of the Ricker stock-recruitment relationship. These results, along with the predator-prey interactions of each species (figure 3), will be used to select species for which a multispecies model should be fitted. Ideally, we want to investigate if the

³Many results in statistics require the vague assumption of ‘large’ n . The general rule-of-thumb of considering $n \geq 30$ as ‘large’ stems from the law of large numbers and the central limit theorem [27]. Our dataset spanning 38 years of stock assessments is therefore traditionally sufficient, but this rule-of-thumb is by no means strict.

⁴Throughout this project, we use $\log$ to mean the natural logarithm.

multispecies model can improve upon the standard model in the case where it already works well for a species, as well as the case where it is not as suitable.

The multispecies model adds additional explanatory variables in the following way:

$$R = aSe^{-bS+cX}, \quad (9)$$

where R and S are the recruitment and stock size respectively of the selected species, and X is the stock size of a predator or prey species. Recall that the recruitment data has been lagged as in section 3.1, so values of both S and X will have been recorded in earlier years than R . If a species has more than one predator or prey, variables are added similarly. Parameter estimates for each added variable should then give insight into how interactions with predators and prey impact the recruitment of model species.

In practice, the non-linear least squares method is applied to a log-transformed version of the model. For example, for the standard Ricker model:

$$\log(R) = \log(aSe^{-bS}) + \log(\epsilon).$$

This is done to ensure the random error ϵ has a multiplicative effect:

$$R = aSe^{-bS} \cdot \epsilon. \quad (10)$$

rather than the additive effect automatically included by the `nls()` function. Multiplicative log-normal errors $\log(\epsilon) \sim N(\mu, \sigma^2)$ are preferred for stock-recruitment models as this prevents any possibility of negative predicted values, which would be biologically impossible. This method will also be used for the multispecies models.

It is important to note that the Ricker function and, by extension, the generalised Ricker function can be linearised:

$$\log\left(\frac{R}{S}\right) = \log(a) - bS + \log(\epsilon), \quad (11)$$

and one may hope to use linear regression techniques to fit such a model, often referred to as transformably linear [25]. However, such a transformation will also affect the assumptions made in the error term. Here, the traditional choice of multiplicative log-normal errors is motivated by biological realism, which must be preserved. For this reason, non-linear regression is applied through the Gauss-Newton method as discussed.

Despite this, the linearised form of the Ricker model still proves useful. Rather than choosing the initial values for the non-linear least squares process, we can calculate optimal starting values by performing linear regression on equation (11) to find estimates for $\log(a)$ and b . By exponentiating the estimate for $\log(a)$, we have the desired improved starting values for estimating a and b . Using these starting values will often result in much fewer iterations of the non-linear least squares algorithm. This idea will also be used to find starting values for the multispecies model using the `lm()` function in R. This process is simplified for the standard Ricker model using the function `srStarts()` from the R package FSA [31].

Because of the error structure of the model, use of the maximum likelihood estimate in calculating the AIC assumes that the model residuals are normally distributed. To verify this, a normal quantile-quantile (Q-Q) plot is used, which plots residuals against quantiles of the standard normal distribution $Z \sim N(0, 1)$. If the points follow a roughly straight line, we can safely assume that the residuals are normally distributed and, therefore, that the AIC is a valid method of comparison.

4 Results and discussion

4.1 Single-species Ricker models

To begin, traditional Ricker models (equation (10)) are fitted for all six species. The resulting models are plotted in figure 5.

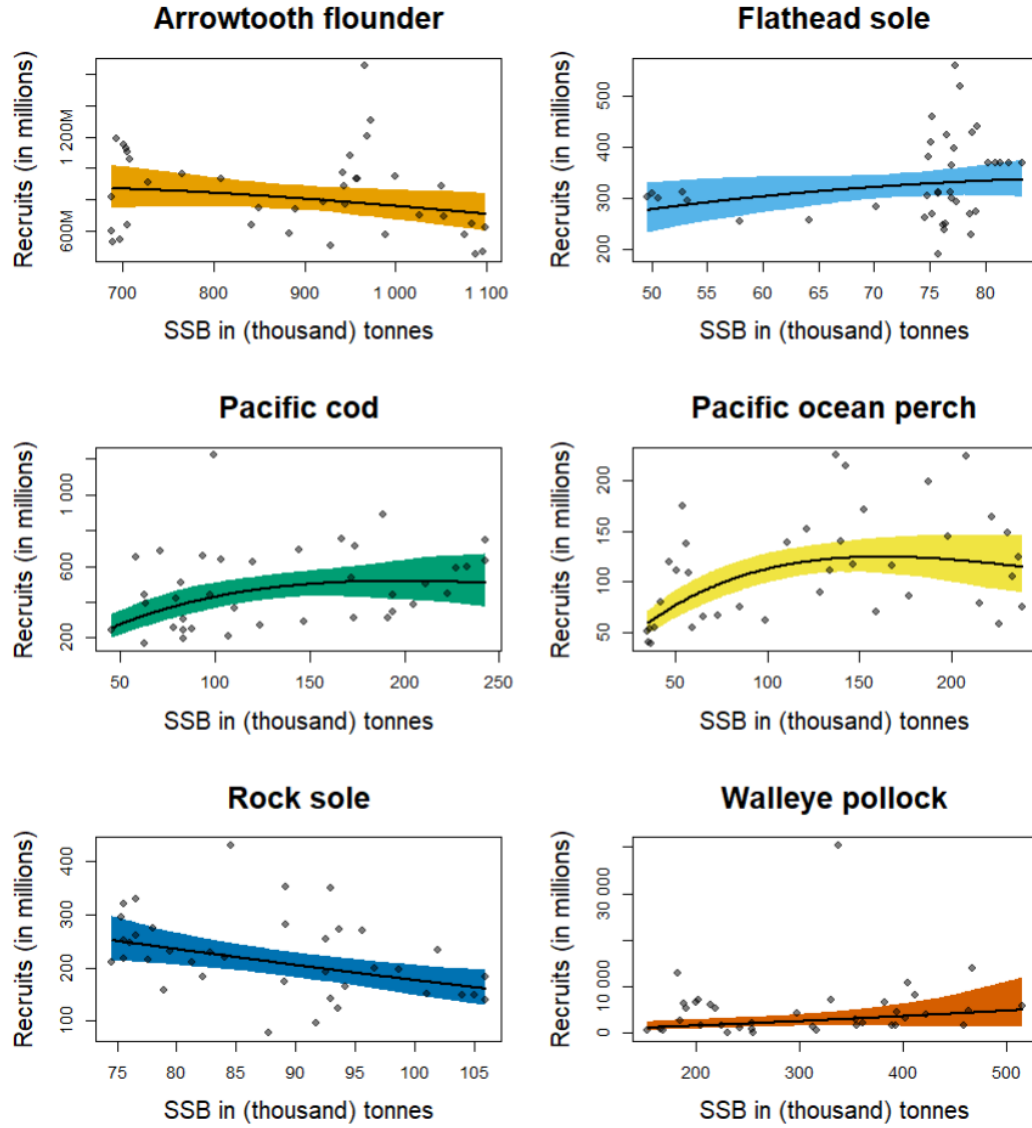


Figure 5: Ricker stock-recruitment curves of six species in the Gulf of Alaska. The coloured region represents the 95% confidence interval of model predictions.

Pacific cod and Pacific ocean perch both demonstrate particularly defined Ricker curves here, whereas the other species appear to have a more flattened curve.

Table 1 displays parameter estimates and confidence intervals for all single species models.

Species	Parameter estimate	95% lower CI	95% upper CI
Arrowtooth flounder	$\hat{a} = 3914.062$ $\hat{b} = 1.642232 \times 10^{-6}$	2165.607 9.752393×10^{-7}	6926.233 2.278479×10^{-6}
Flathead sole	$\hat{a} = 8991.404$ $\hat{b} = 9.582095 \times 10^{-6}$	5327.441 2.587723×10^{-6}	14966.52 1.657656×10^{-5}
Pacific cod	$\hat{a} = 7142.879$ $\hat{b} = 5.075740 \times 10^{-6}$	5095.7 2.860267×10^{-6}	10001.3 7.341443×10^{-6}
Pacific ocean perch	$\hat{a} = 2086.307$ $\hat{b} = 6.166517 \times 10^{-6}$	1607.738 4.345854×10^{-6}	2666.858 8.016725×10^{-6}
Rock sole	$\hat{a} = 22416.4$ $\hat{b} = 2.541815 \times 10^{-5}$	9971.172 1.616062×10^{-5}	57874.08 3.612490×10^{-5}
Walleye pollock	$\hat{a} = 7316.903$ $\hat{b} = -5.645008 \times 10^{-7}$	2966.185 -3.531685×10^{-6}	28705.81 4.671107×10^{-6}

Table 1: Summary of results from the single-species Ricker models.

The Ricker model appears to explain stock-recruitment for Pacific ocean perch particularly well, as its confidence intervals for both a and b are some of the most narrow intervals across all six species. Similarly, the confidence intervals for the parameters in the Pacific cod model also indicate this model is a good fit for the species.

For the flathead sole, the confidence interval for parameter a is wider, suggesting more uncertainty over the number of possible recruits at low stock size. Parameter b sits close to 0, indicating the effect of increased population size on recruitment is minimal.

Looking at the rock sole model, the confidence interval for a is the widest across all species, revealing a clear lack of confidence in possible recruitment at low stock size. Conversely, the confidence interval for b is the most narrow, which provides strong evidence to suggest the effect of increased population size is small ⁵.

Lastly, the walleye pollock model has a wide confidence interval for parameter a . Most notably, the confidence interval for b contains 0, which could suggest that density-dependence provides very little or no information about recruitment ⁶. Note that the unusually strong 2013 recruit cohort of walleye pollock recruits has made it difficult to see the shape of the curve. It would be illogical to simply remove a year from the data, however, to look further into the effect of this data point, a model fitted without using the 2013 data point can be found in the appendix.

⁵For the rock sole, the bootstrapping algorithm did not converge on 17 iterations. This is a very small proportion of the 999 total iterations, so the warnings flagged by R are no cause for concern.

⁶For the walleye pollock, the algorithm did not converge on 105 iterations, likely due to the negative estimate of parameter b . This is a larger proportion ($\approx 10\%$ of all iterations). We continue to use this model for comparison, however this suggests that a standard Ricker curve is not a good fit for walleye pollock.

4.2 Multispecies Ricker models

Now, three multispecies models are presented which incorporate the stock size of interacting species as discussed in section 3.3:

- First, the Pacific ocean perch (section 4.2.1) was selected due to its strong stock-recruitment relationship and relatively simple interaction with only one other species: the walleye pollock.
- Next, the walleye pollock (section 4.2.2) was selected to investigate how such a model would fit for a species with a less defined stock-recruitment relationship and a species involved in many predator-prey interactions.
- Similarly, the Pacific cod (section 4.2.3) was selected due to its many interspecies interactions but with a more defined stock-recruitment relationship.

4.2.1 Pacific ocean perch model

As shown in figure 3, the Pacific ocean perch preys on the walleye pollock and vice versa, so a single additional explanatory variable is added to the Ricker model:

$$R = aSe^{(-bS+cX)},$$

where R and S are the recruitment and stock size respectively of the Pacific ocean perch, and X is the stock size of the walleye pollock. Figure 6 displays the fitted model.

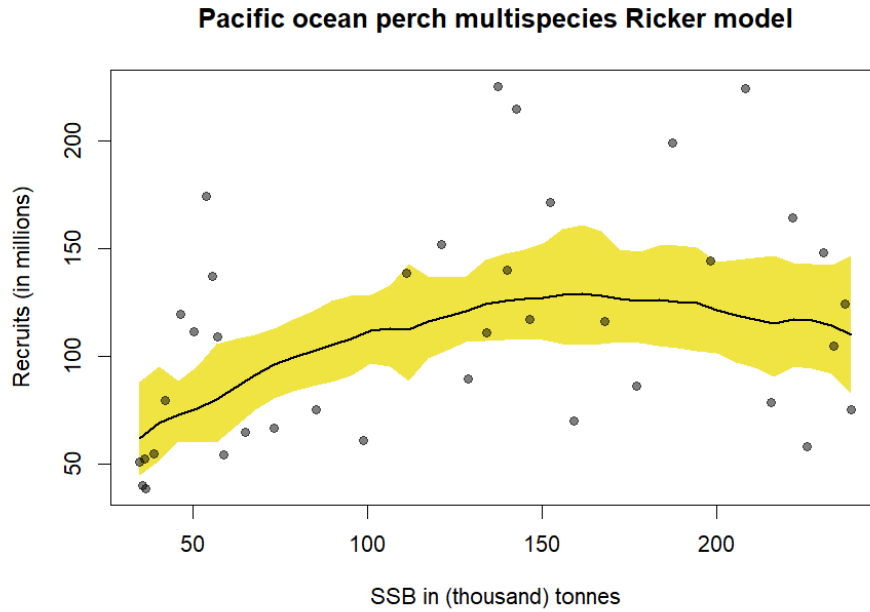


Figure 6: Multispecies Ricker model for Pacific ocean perch incorporating walleye pollock stock size. The yellow region represents the 95% confidence interval of model predictions.

The addition of only a single parameter has kept the model a similar shape to its single-species counterpart. Table 2 summarises some results from parameter estimation ⁷.

Parameter estimate	95% lower CI	95% upper CI
$\hat{a} = 2344$	1260.78	4490.308
$\hat{b} = 6.392 \times 10^{-6}$	4.267315×10^{-6}	8.446968×10^{-6}
$\hat{c} = -2.916 \times 10^{-7}$	-1.741589×10^{-6}	1.0869×10^{-6}

Table 2: Summary of results from the fitted multispecies model for Pacific ocean perch.

The confidence intervals for parameters a and b are relatively narrow. This is consistent with the strong stock-recruitment relationship shown by Pacific ocean perch in the single-species case (see figure 5). The confidence interval for parameter c contains 0, which casts doubt on whether the added parameter has made an improvement on the original model.

The calculated AIC for both the standard and multispecies models are shown in table 3.

Single-species	Multispecies
AIC = 42.6832	AIC = 44.5279

Table 3: Akaike Information Criterion (AIC) of both the single-species and multispecies model for Pacific ocean perch.

The single-species model has a lower AIC than the multispecies model ($42.6832 < 44.5279$), which reinforces a preference for the single-species model for Pacific ocean perch.

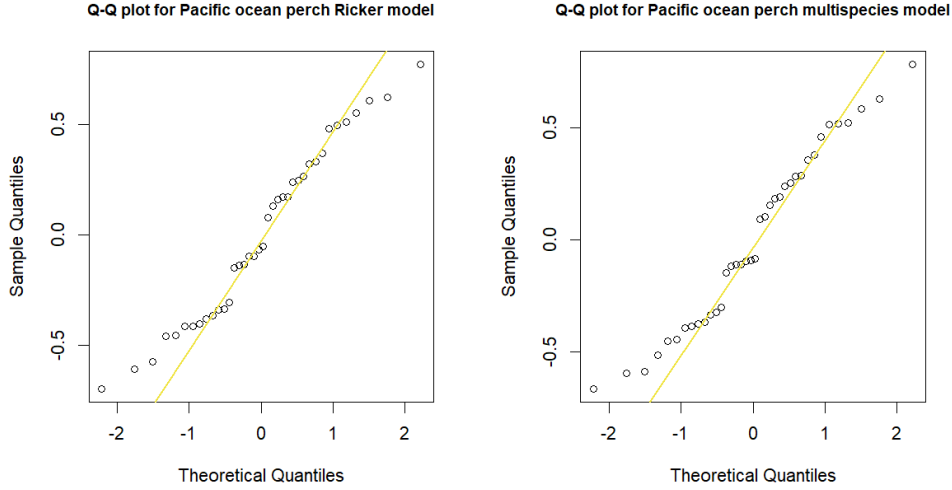


Figure 7: Q-Q plots for standard and multispecies Ricker models for Pacific ocean perch.

⁷Note that here R warns the user that the bootstrapping algorithm did not converge for 7 iterations. As before, this is a very small number of the total iterations and so this is not indicative of any larger issues.

The Q-Q plots in figure 7 suggest normal residuals in each model, and therefore the assumptions made in calculating the AIC have been satisfied.

4.2.2 Walleye pollock model

Again referring to figure 3, the walleye pollock is preyed on by all 5 other species and is a predator of the flathead sole, Pacific cod and Pacific ocean perch. Here, 5 additional explanatory variables are added to form the 7 parameter model:

$$R = aSe^{(-bS+cX_1+dX_2+fX_3+gX_4+hX_5)},$$

where R and S are the recruitment and stock size of the walleye pollock, and X_{1-5} are the stock sizes of the arrowtooth flounder, flathead sole, Pacific cod, Pacific ocean perch and rock sole respectively. Figure 8 shows the fitted model.

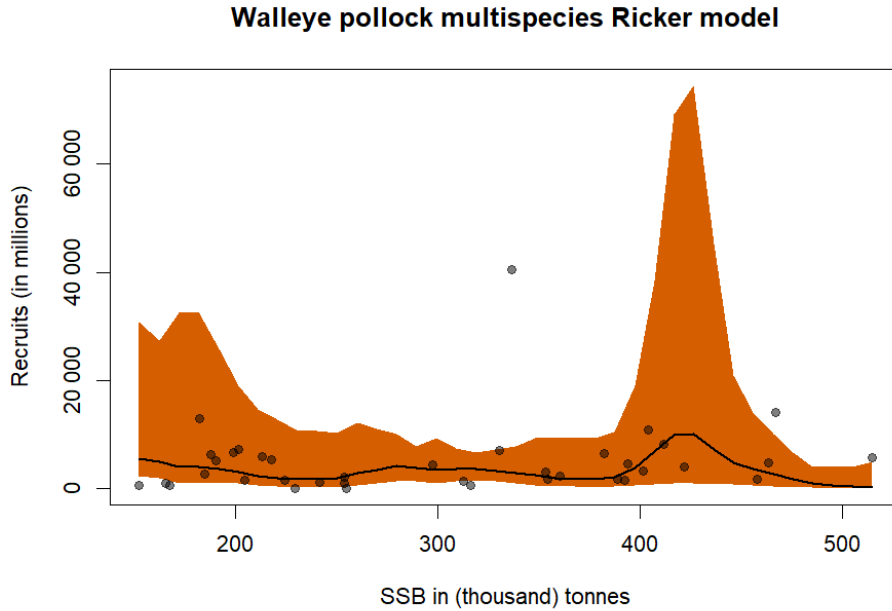


Figure 8: Multispecies Ricker model for walleye pollock incorporating stock sizes of interacting species. The orange region represents the 95% confidence interval of model predictions.

The effect of the large 2013 recruit cohort is clearly present in figure 8. Table 4 contains the results of parameter estimation for the model ⁸.

⁸The negative parameters have again created issues in bootstrapping, as in this case there were 376 unsuccessful iterations. This is a clear sign that this model is also unsuitable, a fact that is backed up by the parameter confidence intervals.

Parameter estimate	95% lower CI	95% upper CI
$\hat{a} = 1754$	695.33	469532.5
$\hat{b} = 2.782 \times 10^{-7}$	-6.775137×10^{-6}	8.261399×10^{-6}
$\hat{c} = 5.573 \times 10^{-6}$	-1.595753×10^{-6}	1.050025×10^{-5}
$\hat{d} = -7.517 \times 10^{-5}$	-1.690138×10^{-4}	4.877814×10^{-7}
$\hat{f} = -1.101 \times 10^{-5}$	-4.308922×10^{-5}	1.727712×10^{-5}
$\hat{g} = -1.713 \times 10^{-5}$	-3.470147×10^{-5}	5.483864×10^{-6}
$\hat{h} = 6.647 \times 10^{-5}$	-1.986017×10^{-5}	1.635078×10^{-4}

Table 4: Summary of results from the fitted multispecies model for walleye pollock.

Here, parameter a has an exceptionally wide confidence interval, suggesting this model tells us very little about how many recruits to expect at low stock size. Confidence intervals for all other model parameters contain 0. For b , this is consistent with the single-species model (see table 1). For the added parameters, this unfortunately suggests incorporating stock size of other species in this way may not provide any additional information about walleye pollock recruitment.

Table 5 displays the calculated AIC for both the single-species and multispecies models.

Single-species	Multispecies
AIC = 145.0259	AIC = 138.1454

Table 5: Akaike Information Criterion (AIC) of both the single-species and multispecies model for walleye pollock.

In this case, although containing 5 additional parameters, the multispecies model has a slightly lower AIC than the single-species model ($138.1454 < 145.0259$), indicating a weak preference for the multispecies model for walleye pollock.

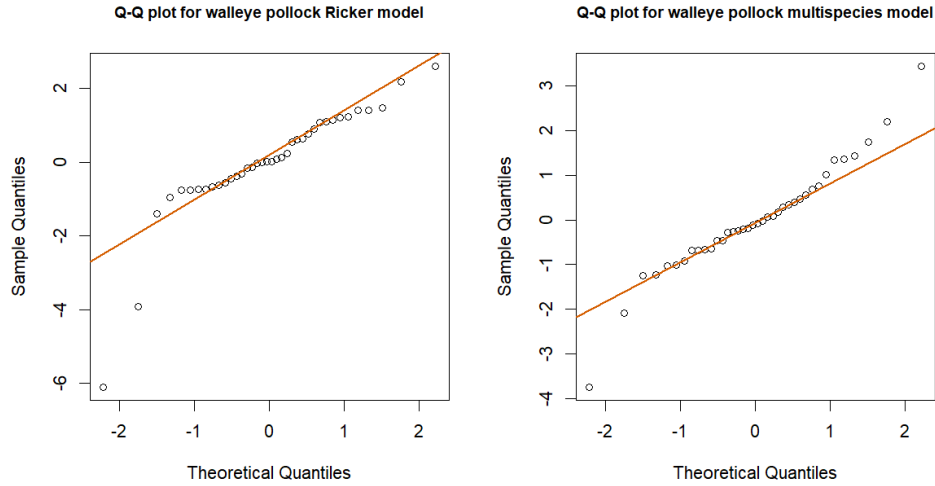


Figure 9: Q-Q plots for standard and multispecies Ricker models for walleye pollock.

Figure 9 indicates relatively normal residuals, which satisfies the assumption made in using the AIC.

4.2.3 Pacific cod model

Referring back to figure 3, the Pacific cod is a predator of the arrowtooth flounder, flathead sole, rock sole, and walleye pollock, with the latter species also a predator of Pacific cod. Four additional explanatory variables are therefore added to the model:

$$R = aSe^{(-bS+cX_1+dX_2+fX_3+gX_4)},$$

where R and S are the recruitment and stock size of the Pacific cod, and X_{1-4} are the stock sizes of the arrowtooth flounder, flathead sole, rock sole and walleye pollock respectively. The fitted model is shown in figure 10.

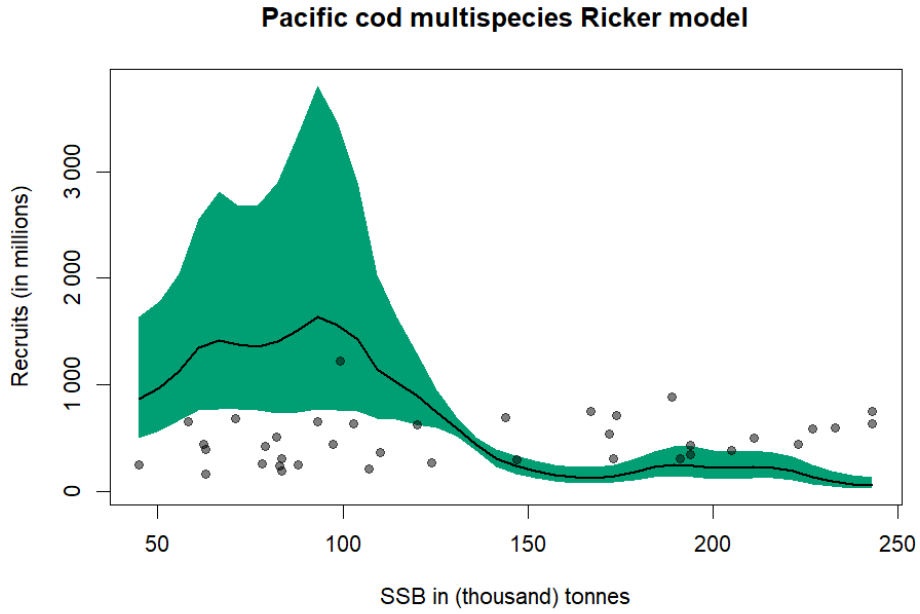


Figure 10: Multispecies Ricker model for Pacific cod incorporating stock sizes of interacting species. The green region represents the 95% confidence interval of model predictions.

The Ricker curve here appears to have been highly influenced by the inclusion of predator and prey stock size, particularly at high population sizes.

The results of parameter estimation can be found in table 6 ⁹.

Parameter estimate	95% lower CI	95% upper CI
$\hat{a} = 10240$	4456.93	35435.57
$\hat{b} = 1.475 \times 10^{-5}$	9.94242×10^{-6}	1.99773×10^{-5}
$\hat{c} = -7.954 \times 10^{-7}$	-2.606904×10^{-6}	7.095725×10^{-7}
$\hat{d} = -4.815 \times 10^{-5}$	-6.516048×10^{-5}	-3.171604×10^{-5}
$\hat{f} = 5.536 \times 10^{-5}$	3.395207×10^{-5}	7.788213×10^{-5}
$\hat{g} = 9.461 \times 10^{-7}$	-6.616602×10^{-7}	2.487671×10^{-6}

Table 6: Summary of results from the fitted multispecies model for Pacific cod.

The confidence intervals for a and b are both wider compared to the single-species case (see table 1), indicating additional parameters have increased uncertainty over the standard density-dependent and independent effects.

The confidence intervals for parameters c and g , associated with the stock size of arrowtooth flounder and walleye pollock respectively, both contain 0. This puts into question the value of their contribution to the model.

On the contrary, parameters d and f are unlikely to be 0, which suggests the addition of the stock size of flathead sole and rock sole respectively is impacting the model. From a mathematician's perspective, this small success motivates fitting a model for Pacific cod with only these species added. However, it should be noted that juvenile walleye pollock are one of the main sources of food for adult Pacific cod [32]. This fact casts some doubt on the validity of the model, and any model incorporating only flathead sole and rock sole would stray further from ecological realism.

Single-species	Multispecies
AIC = 50.6826	AIC = 30.50261

Table 7: Akaike Information Criterion (AIC) of both the single-species and multispecies model for Pacific cod.

Finally, despite four additional parameters, the multispecies model has a much lower AIC than the single-species model, suggesting a preference for the multispecies model for Pacific cod.

⁹Iterations fail to converge 65 times during bootstrapping. Again, this is a small proportion of the total number of iterations, so any warnings brought up by R are not serious causes for concern.

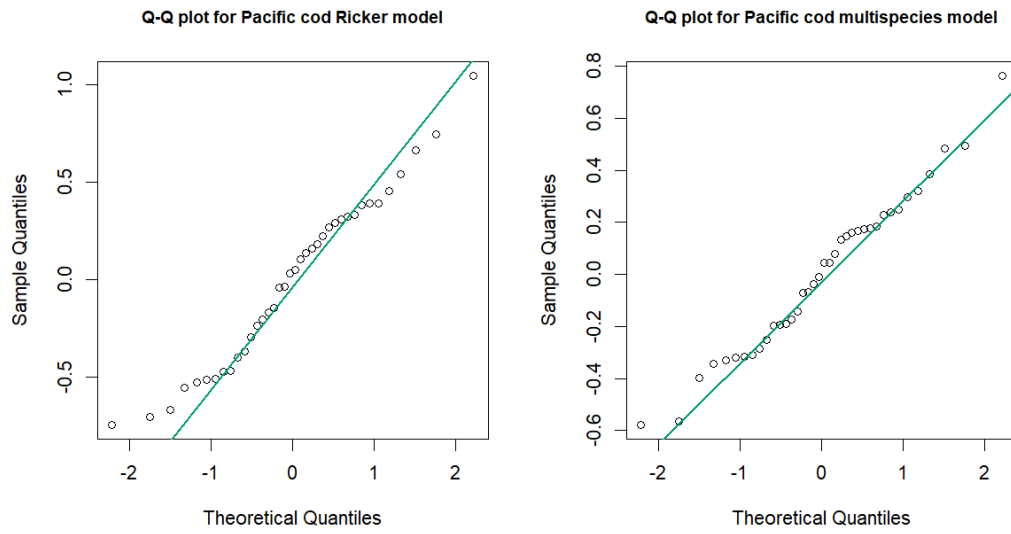


Figure 11: Q-Q plots for standard and multispecies Ricker models for Pacific cod.

Again, the residuals lie around the straight line in figure 11, so the assumption made in calculating the AIC has been satisfied.

5 Conclusion

5.1 Summary

The recruitment process, by which young fish transition to an older life stage, is known to be related to the size of the fish stock [11]. This relationship can be modelled mathematically, which can produce predictions about the population and, in turn, can inform management decisions in the fishery. However, such a relationship can often be challenging to describe, as stock-recruitment data often seems random with no clear trend, leading stock-recruitment modelling to be used only when the data appears suitable. Attempts have been made to improve the fit of stock-recruitment models by including additional explanatory variables, such as surface sea temperature [16], and here, this method was adapted to instead incorporate the stock size of predator and prey species.

The selected stock-recruitment model was proposed by Ricker [15]:

$$R = \alpha S e^{-\beta S},$$

where S is the stock size and R is the number of recruits in a given year. This model was generalised to include an additional explanatory variable X :

$$R = a S e^{(-bS+cX)}.$$

To incorporate multispecies effects, here X is the stock size of an interacting predator or prey species of the species we wish to model. Additional species are added similarly.

Data for this project was selected from the RAM Legacy Stock Assessment Database [17], where we searched for a suitable collection of species in a single fishing region. The Gulf of Alaska was selected, where six interacting species had stock size and recruitment time series available: arrowtooth flounder, flathead sole, Pacific cod, Pacific ocean perch, rock sole and walleye pollock.

Both the standard models and their multispecies were fitted to the data and interpreted using the statistical software R [29], following the methods of Ogle [30]. Non-linear least squares regression [25] was applied to estimate model parameters, which were then interpreted through confidence intervals calculated using the bootstrap sampling method [26]. Finally, the multispecies models were compared to their single-species equivalents using the Akaike Information Criterion [28].

Among the standard Ricker models for the six species, Pacific ocean perch and Pacific cod were best suited to this type of stock-recruitment relationship. However, the parameter estimates for walleye pollock cast doubt on the inclusion of the density-dependent effect represented by parameter b .

Three multispecies models were fitted, which provided varied results. Firstly, for the Pacific ocean perch, a narrow confidence interval about 0 for the added parameter c , representing the stock size of walleye pollock, suggested the additional variable may not have improved the original model. This interpretation was reinforced by a lower AIC for the single-species model compared with that of the multispecies model.

Then, for the walleye pollock, it was hoped consideration of the five predator and prey species would improve the model fit. A weak preference for the multispecies model was indicated by the calculated AIC. However, the density-dependent effect was again questioned,

as well as the contribution of every added parameter.

Lastly, for the Pacific cod, four predator-prey species were included. As with the other models, the presence of 0 in parameter confidence intervals for the stock size of arrowtooth flounder and walleye pollock indicates their addition did not improve the model fit. However, this contradicts the biological fact that juvenile walleye pollock comprise a large proportion of the diet of Pacific cod [32]. Interestingly, the confidence intervals for the parameters associated with flathead sole and rock sole did not contain 0, indicating their presence had impacted the model. Likewise, the AIC of the multispecies model was significantly lower than that of the single-species model. Together, these results indicate that this type of model could more accurately describe stock-recruitment for some species.

5.2 Project limitations and future work

The methodology in this project has some limitations which could be explored further. Through the choice of a specific dataset of six species, this methodology could be at risk of data dredging, where careless selection and elimination of data can lead to false results. However, here we have prevented this initially through data selection guided by Alaskan ecology, and also by interpreting confidence intervals for parameter estimates in place of hypothesis tests, which are at increased risk of false positives under data dredging.

Other types of stock-recruitment relationship could be chosen: here the Ricker model is chosen purely as it has a well-established extension to additional variables, but we saw that the standard Ricker model was not an appropriate fit for all species. Additional explanatory variables could be added to other models, for example, the Shepherd stock-recruitment model [33]:

$$R = \frac{\alpha S}{1 + \beta S^\gamma},$$

which itself generalises the Beverton-Holt model, obtained precisely when $\gamma = 1$, and the dome shape of the Ricker model, obtained when $\gamma > 1$ ¹⁰. This model is likely a better fit for a wider array of fish species, but the inclusion of parameter γ would require further estimation and subsequently increase the complexity of the developed multispecies Shepherd model.

Here, we focused on predator-prey interactions, although competition for resources has also been indirectly explored, if prey abundance is viewed as a dietary resource, particularly in the case of walleye pollock. Competition could be investigated further by incorporating the amount of a resource and the population size of species that share that resource. Depending on region and species, these could be food resources besides juvenile fish, such as algae or plankton, or instead the available space in a shared habitat, which could be explored through the total abundance of all marine species in a given area. The latter would be of particular interest if the size of the habitat in question has changed over time, as researchers and fishermen have seen in coral reef habitats in recent years [35].

It is also important to note that our choice of data in this project was limited to managed fish stocks, while many more species exist in the Gulf of Alaska that aren't monitored

¹⁰Note when $\gamma < 1$, the Shepherd model shares the characteristics of the Cushing model $R = \alpha S^\gamma$ [34]. This model is very rarely used in practice, as it lacks a limit for recruitment: infinite spawning stock would lead to infinite recruitment.

by fisheries. To maximise the accuracy of any model accounting for species interactions, we must have data on the population size of all known predator and prey species for the managed species of interest. This process could be aided by careful analysis of fish unintentionally caught when fishing for a particular species, referred to as bycatch. However, recent conservation concerns have led to a change in harvesting methods to mitigate unsustainable amounts of bycatch [36]. Because of this, it is likely that such data would need to be collected with a model in mind.

As an extension of typical stock-recruitment models, the simplicity of this methodology does lend itself well to stock assessments. However, if all the improvements discussed here are implemented, that simplicity would be lost, and fishery scientists would face more work in devising each year's stock assessment and communicating results with confidence. The importance of how real people use mathematical research should not be understated, particularly in fisheries management, as sustainably managed stocks will go on to provide food for themselves, their peers and millions of others around the world.

Appendix - an unusual year for walleye pollock

In section 3.1, we note that the unusually large 2012 year class of walleye pollock caused a large spike in recruitment when they reached age 1 in 2013. We see later in section 4.1 that this data point has a dominating effect on the fit of the standard Ricker model. The analysis continues to use the data point as it could help indicate the effect of species interaction on recruitment in the multispecies models. Of course, in reality fisheries scientists would not exclude a year of data simply for convenience. In this appendix, a standard Ricker model is fitted without considering the 2012 year class to further investigate its effect.

Recall the Ricker stock-recruitment model (equation (7)):

$$R = aSe^{-bS},$$

where R is the number of walleye pollock recruits, S is the size of the walleye pollock stock, measured as spawning stock biomass (SSB), and a and b are parameters to be estimated. Figure 12 displays the fitted model.

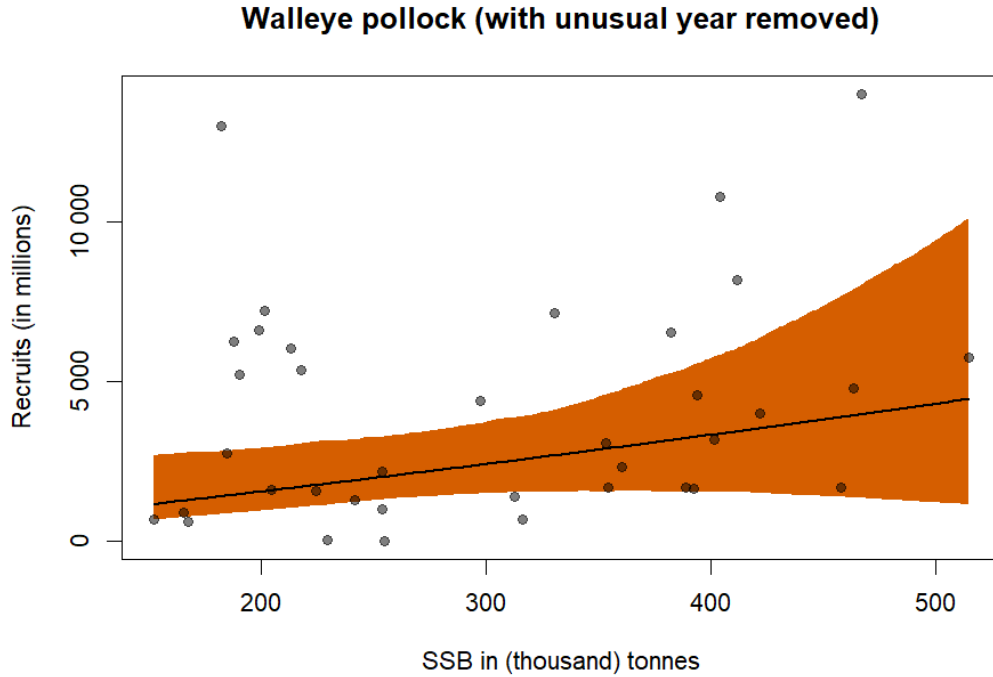


Figure 12: Ricker stock-recruitment model for walleye pollock, excluding the unusual 2012 year class. The orange region represents the 95% confidence interval of model predictions.

We can now see more clearly the shape of the curve, which appears to be quite flat and increasing with no peak in recruitment. This latter observation indicates that perhaps the Ricker model, with its characteristic dome shape (see figure 1), is not the best choice of stock-recruitment relationship for walleye pollock.

Parameter estimates and confidence intervals for this model are shown in table 8 ¹¹ .

Parameter estimate	95% lower CI	95% upper CI
$\hat{a} = 7323.056$	3049.401	33572.37
$\hat{b} = -3.27947 \times 10^{-7}$	-3.085087×10^{-6}	4.621645×10^{-6}

Table 8: Summary of results from the Ricker model for walleye pollock excluding the unusual 2012 year class.

We can draw many similarities between the original model and this updated model, indicating that the unusual data point does not have much of an impact on the model fit. The estimate for parameter a has increased by ≈ 6 , and its confidence interval has widened further, which we can expect with fewer data points. The estimate for parameter b has also increased slightly, but the confidence interval remains similarly narrow and still contains 0, reinforcing our doubts about the importance of density-dependence in walleye pollock recruitment.

¹¹As with the original walleye pollock model, bootstrapping causes R to raise some warnings. The algorithm fails to converge on 87 iterations, a slightly smaller proportion ($\approx 9\%$) of all iterations. This is again likely due to the negative estimate of parameter b and reinforces the idea that a Ricker curve is not a good fit here.

References

- [1] Food and Agriculture Organization of the United Nations. *FAO Report: Global fisheries and aquaculture production reaches a new record high*. 6th July 2024. URL: <https://www.fao.org/newsroom/detail/fao-report-global-fisheries-and-aquaculture-production-reaches-a-new-record-high/en>.
- [2] FAO et al. *The State of Food Security and Nutrition in the World 2025 - Addressing high food price inflation for food security and nutrition*. FAO; IFAD; UNICEF; WFP; WHO; 27th July 2025. DOI: 10.4060/cd6008en.
- [3] Kashif Abbass et al. “A review of the global climate change impacts, adaptation, and sustainable mitigation measures”. In: *Environmental Science and Pollution Research* 29.28 (1st June 2022), pp. 42539–42559. ISSN: 1614-7499. DOI: 10.1007/s11356-022-19718-6.
- [4] Cole C. Monahan et al. *Assessment of the Walleye Pollock Stock in the Gulf of Alaska*. 2024. URL: <https://www.npfmc.org/library/safe-reports/>.
- [5] A J R Cotter et al. “Are stock assessment methods too complicated?” In: *Fish and Fisheries* 5.3 (2004), pp. 235–254. ISSN: 1467-2979. DOI: 10.1111/j.1467-2679.2004.00157.x.
- [6] A. W. Stoner. “Effects of environmental variables on fish feeding ecology: implications for the performance of baited fishing gear and stock assessment”. In: *Journal of Fish Biology* 65.6 (2004), pp. 1445–1471. ISSN: 1095-8649. DOI: 10.1111/j.0022-1112.2004.00593.x.
- [7] H. O. Pörtner and M. A. Peck. “Climate change effects on fishes and fisheries: towards a cause-and-effect understanding”. In: *Journal of Fish Biology* 77.8 (2010), pp. 1745–1779. ISSN: 1095-8649. DOI: 10.1111/j.1095-8649.2010.02783.x.
- [8] Kiersten L. Curti et al. “Evaluating the performance of a multispecies statistical catch-at-age model”. In: *Canadian Journal of Fisheries and Aquatic Sciences* 70.3 (Mar. 2013), pp. 470–484. ISSN: 0706-652X. DOI: 10.1139/cjfas-2012-0229.
- [9] Villy Christensen and Carl J Walters. “Ecopath with Ecosim: methods, capabilities and limitations”. In: *Ecological Modelling*. Placing Fisheries in their Ecosystem Context 172.2 (1st Mar. 2004), pp. 109–139. ISSN: 0304-3800. DOI: 10.1016/j.ecolmodel.2003.09.003.
- [10] Ray Hilborn and Carl J. Walters. *Quantitative Fisheries Stock Assessment: Choice, Dynamics and Uncertainty*. Springer Science & Business Media, 1992.
- [11] Ed Camp et al. “Fish Population Recruitment: What recruitment means and why it matters: FA222, 3/2020”. In: *EDIS* 2020.2 (27th Mar. 2020), pp. 6–6. ISSN: 2576-0009. DOI: 10.32473/edis-fa222-2020.
- [12] Yingxi Yang and Takashi Yamakawa. “Re-examination of stock–recruitment relationships: a meta-analysis”. In: *ICES Journal of Marine Science* 79.4 (1st May 2022), pp. 1380–1393. ISSN: 1054-3139. DOI: 10.1093/icesjms/fsac061.
- [13] Terrance J. Quinn and Richard B. Deriso. *Quantitative Fish Dynamics*. Oxford University Press, 25th Mar. 1999. 561 pp. ISBN: 978-0-19-536040-0.
- [14] Raymond J. H. Beverton and Sidney J. Holt. *On the Dynamics of Exploited Fish Populations*. Springer Science & Business Media, 1957. 541 pp. ISBN: 978-94-011-2106-4.

- [15] W. E. Ricker. “Stock and Recruitment”. In: *Journal of the Fisheries Research Board of Canada* 11.5 (May 1954), pp. 559–623. ISSN: 0015-296X. DOI: 10.1139/f54-039.
- [16] Geir Ottersen et al. “Temporal shifts in recruitment dynamics of North Atlantic fish stocks: effects of spawning stock and temperature”. In: *Marine Ecology Progress Series* 480 (22nd Apr. 2013), pp. 205–225. ISSN: 1616-1599, 0171-8630. DOI: 10.3354/meps10249.
- [17] Daniel Ricard et al. “Examining the knowledge base and status of commercially exploited marine species with the RAM Legacy Stock Assessment Database”. In: *Fish and Fisheries* 13.4 (2012), pp. 380–398. ISSN: 1467-2979. DOI: 10.1111/j.1467-2979.2011.00435.x.
- [18] North Pacific Fishery Management Council. *Fishery Management Plan for the Groundfish fisheries of the Gulf of Alaska*. Oct. 2024. URL: <https://www.fisheries.noaa.gov/management-plan/groundfish-gulf-alaska-management-plan>.
- [19] Rainer Froese and Daniel Pauly. *FishBase*. 2025. URL: <https://www.fishbase.org/>.
- [20] Mei-Sun Yang and Mark W. Nelson. *Food habits of the commercially important groundfishes in the Gulf of Alaska in 1990, 1993, and 1996*. 2000. URL: <https://repository.library.noaa.gov>.
- [21] P.A. Livingston. “Importance of predation by groundfish, marine mammals and birds on walleye pollock *Theragra chalcogramma* and Pacific herring *Clupea pallasii* in the eastern Bering Sea”. In: *Marine Ecology Progress Series* (30th Dec. 1993).
- [22] Richard D. Brodeur and Patricia A. Livingston. *Food habits and diet overlap of various eastern Bering Sea fishes*. 1988. URL: https://repository.library.noaa.gov/view/noaa/5781/noaa_5781_DS1.pdf.
- [23] National Oceanic and Atmospheric Administration (NOAA) Fisheries. *Fish species illustrations*. 21st Mar. 2025. URL: <https://www.fisheries.noaa.gov/fish-sharks#by-species> (visited on 06/03/2026).
- [24] Ludwig Fahrmeir et al. *Regression: Models, Methods and Applications*. Springer, 2013. ISBN: 978-3-642-34332-2 978-3-642-34333-9. DOI: 10.1007/978-3-642-34333-9.
- [25] Bates, D.M. and Watts, D.G. *Nonlinear Regression Analysis and its Applications*. New York: Wiley, 1988. ISBN: 0-471-81643-4.
- [26] S.Huet et al. *Statistical Tools for Nonlinear Regression: A Practical Guide with S-PLUS and R Examples*. 2nd. Springer Series in Statistics. Springer-Verlag, 2004. ISBN: 978-0-387-40081-5. DOI: 10.1007/b97288.
- [27] Frederik Michel Dekking et al. *A Modern Introduction to Probability and Statistics: Understanding Why and How*. Springer Texts in Statistics. Springer, 2005. ISBN: 978-1-85233-896-1 978-1-84628-168-6. DOI: 10.1007/1-84628-168-7.
- [28] Hirotugu Akaike. “Information Theory and an Extension of the Maximum Likelihood Principle”. In: *Selected Papers of Hirotugu Akaike*. Ed. by Emanuel Parzen, Kunio Tanabe and Genshiro Kitagawa. New York: Springer, 1973, pp. 199–213. ISBN: 978-1-4612-1694-0. DOI: 10.1007/978-1-4612-1694-0_15.
- [29] R Core Team. *R: A Language and Environment for Statistical Computing*. 2024. URL: <https://www.r-project.org/>.
- [30] Derek H. Ogle. *Introductory Fisheries Analyses with R*. New York: Chapman and Hall/CRC, 3rd Sept. 2018. 337 pp. ISBN: 978-1-315-37198-6. DOI: 10.1201/9781315371986.

- [31] Derek H. Ogle et al. *FSA: Simple Fisheries Stock Assessment Methods*. Version 0.10.1. 10th Jan. 2026.
- [32] Daniel Urban. “Food habits of Pacific cod and walleye pollock in the northern Gulf of Alaska”. In: *Marine Ecology Progress Series* 469 (2012), pp. 215–222. ISSN: 0171-8630.
- [33] J. G. Shepherd. “A versatile new stock-recruitment relationship for fisheries, and the construction of sustainable yield curves”. In: *Journal du Conseil* 40.1 (1st Feb. 1982), pp. 67–75. ISSN: 0020-6466. DOI: 10.1093/icesjms/40.1.67.
- [34] D. H. Cushing. “The Dependence of Recruitment on Parent Stock in Different Groups of Fishes”. In: *Journal du Conseil* 33.3 (1st May 1971), pp. 340–362. ISSN: 0020-6466. DOI: 10.1093/icesjms/33.3.340.
- [35] Morgan S. Pratchett et al. “Changes in Biodiversity and Functioning of Reef Fish Assemblages following Coral Bleaching and Coral Loss”. In: *Diversity* 3.3 (Sept. 2011), pp. 424–452. ISSN: 1424-2818. DOI: 10.3390/d3030424.
- [36] Lisa M. Komoroske and Rebecca L. Lewison. “Addressing fisheries bycatch in a changing world”. In: *Frontiers in Marine Science* 2 (27th Oct. 2015). ISSN: 2296-7745. DOI: 10.3389/fmars.2015.00083.